DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

First Semester, SY 2024-2025

COURSE CODE	:	S-ITPE004LA
COURSE TITLE	:	IT PROFESSIONAL ELECTIVE 4 (SYSTEM INTEGRATION AND ARCHITECTURE 2 LAB)
COURSE TYPE	:	Laboratory
COURSE CREDIT	:	One Unit
LEARNING MODALITY	:	Fully Onsite
PRE-REQUISITES	:	SYSTEM INTEGRATION & ARCHITECTURE 1(LEC & LAB)
CO-REQUISITES	:	S-ITPE004
PROFESSOR	:	MARYLI F. ROSAS, DIT
CONSULTATION HOURS	:	TUES 10:00-12:00 ICT209 WED 10:00-12:00 ICT209 FRI 7:00-10:00 & 1:00-2:30 MS Teams
CONTACT INFORMATON	:	mfrosas@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course focuses on project management, testing and quality assurance, and organizational context architecture. The course introduces the students on usability testing and recommends appropriate actions based on the results of the test. Furthermore, topics on the construction of an architectural model of a complex system using an architectural framework are discussed.

This course builds upon the fundamentals of system architecture and integration, exploring advanced concepts and techniques for designing, integrating, and managing complex systems. Emphasis is placed on real-world applications, case studies, and hands-on projects.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO 1. evaluate software requirements and appraise the processes involved in discovering and documenting these requirements;
CLO 2. examine different system models that may be developed as part of requirements engineering and system design processes;
CLO 3. analyze the concepts of software architecture and architectural design;
CLO 4. evaluate object-oriented software design using the UML and highlight important implementation concerns;
CLO 5. identify software testing and software testing processes;
CLO 6. explain why software evolution is such an important part of software engineering and to describe the challenges of maintaining a large base of software systems, developed over many years;
CLO 7. discuss software project management and two important management activities, namely, risk management and people management; and
CLO 8. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 1. Evaluate software requirements and appraise the processes involved in discovering and documenting these requirements.	• TLO 1. Evaluate the concepts of user and system requirements and why these requirements should be written in different ways. • TLO 2. Value the differences between functional and nonfunctional software requirements. • TLO 3. Appraise the main requirements engineering activities of elicitation, analysis, and validation, and the relationships between these activities. • TLO 4. Weigh why requirements management is necessary and how it supports other requirements engineering activities.
CLO 2. Examine different system models that may be developed as part of requirements engineering and system design processes.	• TLO 5. Relate how graphical models can be used to represent software systems and why several types of models are needed to fully represent a system. • TLO 6. Analyze the fundamental system modeling perspectives of context, interaction, structure, and behavior. • TLO 7. Distinguish the principal diagram types in the Unified Modeling Language (UML) and how these diagrams may be used in system modeling. • TLO 8. Relate to model-driven engineering, where an executable system is automatically generated from structural and behavioral models.

CLO 3. Analyze the concepts of software architecture and architectural design.	• TLO 9. Distinguish why the architectural design of software is important. • TLO 10. Examine the decisions that must be made about the software architecture during the architectural design process. • TLO 11. Relate to the idea of Architectural patterns, well-tried ways of organizing software architectures that can be reused in system designs. • TLO 12. Distinguish how Application-Specific Architectural patterns may be used in transaction processing and language processing systems.
CLO 4. Evaluate object-oriented software design using the UML and highlight important implementation concerns.	• TLO 13. Appraise the most important activities in a general, object-oriented design process. • TLO 14. Select some of the different models that may be used to document an object-oriented design. • TLO 15. Support the idea of design patterns and how these are a way of reusing design knowledge and experience. • TLO 16. Weigh key issues that must be considered when implementing software, including software reuse and open-source development.
CLO 8. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Allotted Hours**
Module 1 TLO 1 TLO 2 TLO 3 TLO 4 CLO 1 CLO 8	Week 1 Aug 19-24	• Gospel Reading, Reflection, and Prayer • Syllabus uploaded. • Presentation of course introduction • Discussion of Module 1 – Requirements Engineering EA: : Lab Activity 1: • Brainstorming / Checking / Approval of Topics for Project (By Group) Project Grouping (3 members per group)	MS Teams PowerPoint Schoolbook	3
	Week 2 Aug 26-31	• Gospel Reading, Reflection, and Prayer • Continuation of Discussion of Module 1 – Requirements Engineering • EA: Lab Activity 2: Answer end of chapter exercises on page 41, by group	MS Teams PowerPoint Schoolbook	3

Module 2 TLO 5 TLO 6 TLO 7 TLO 8 CLO 2 CLO 8	Week 3 Sept 2-7	• Gospel Reading, Reflection, and Prayer Discussion of Module 2 – Software Processes Enabling Assessment: Lab Activity 3: By group, answer the following question based on your approved topic: Suggest the most appropriate generic software process model that might be used as a basis for managing the development of your group project. Explain your answer according to the type of system your group is developing.	MS Teams PowerPoint Schoolbook	3
	Week 4 Sept 9-14	• Gospel Reading, Reflection, and Prayer • Discussion of Module 3 – Agile Software Development	MS Teams PowerPoint Schoolbook	3
	Week 5 Sept 16-21	• Gospel Reading, Reflection, and Prayer • Discussion of Module 3 – Architectural Design • EA: Submission of Initial Progress of Group Project's Documentation .	MS Teams PowerPoint Schoolbook	3
Module 3 Module 4 TLO 9 TLO 10 TLO 11 TLO 12 TLO 13 TLO 14 TLO15 TLO 16 CLO 3 CLO 4 CLO 8	Week 6 Sept 23-28	• Gospel Reading, Reflection, and Prayer Continuation of Discussion of Module 3 – Agile Software Development EA: Lab Activity 4: Consider the following scenario and answer the question at the end. Imagine that a government wants a software program that helps to keep track of the utilization of the country's vast mineral resources. Although the requirements put forward by the government were not very clear, a software company was tasked with the development of a prototype. The government found the prototype impressive, and asked it be extended to be the actual system that would be used. Discuss the pros and cons of taking this approach. This activity is to be performed individually.	MS Teams PowerPoint Schoolbook	3
	Week 7 Sept 30 - Oct 5	• Gospel Reading, Reflection, and Prayer • Discussion of Module 4 – Requirements Engineering Lab Activity 5: Explain how the principles underlying agile methods lead to the accelerated development and deployment of software, in relation to your group project. This activity must be answered by group.	MS Teams PowerPoint Schoolbook	3

	Week 8 Oct 7-12	• Gospel Reading, Reflection, and Prayer • Checking of Thesis Progress Report (Software and Documentation) • Summative Examination	PowerPoint Schoolbook	3
TOTAL				24

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 5. Comprehend software testing and software testing processes.	• TLO 17. Understand the stages of testing from testing during development to acceptance testing by system customers. • TLO 18. Identify techniques that help us choose test cases that are geared to discovering program defects. • TLO 19. Recognize test-first development, where you design tests before writing code and run these tests automatically. • TLO 20. Discuss the three distinct types of testing—component testing, system testing, and release testing • TLO 21. Explain the distinctions between development testing and user testing.
CLO 6. Explain why software evolution is such an important part of software engineering and to describe the challenges of maintaining a large base of software systems, developed over many years.	• TLO 22. Recognize that software systems must adapt and evolve if they are to remain useful and that software change and evolution should be considered as an integral part of software engineering. • TLO 23. Discuss what is meant by legacy systems and why these systems are important to businesses. • TLO 24. Describe how legacy systems can be assessed to decide whether they should be scrapped, maintained, reengineered, or replaced. • TLO 25. Recognize different types of software maintenance and the factors that affect the costs of making changes to legacy software systems.
CLO 7. Discuss software project management and two important management activities, namely, risk management and people management.	• TLO 26. Define the principal tasks of software project managers. • TLO 27. Explain risk management and some of the risks that can arise in software projects. • TLO 28. Discuss factors that influence personal motivation and what these might mean for software project managers. • TLO 29. Recognize key issues that influence team working, such as team composition, organization, and communication.
CLO 8. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Allotted Hours
Module 5 Module 6 TLO 17 TLO 18 TLO 19 TLO 20 TLO 21 TLO 22 TLO 23 TLO 24 TLO 25 CLO 5 CLO 6 CLO 8	Week 10 Oct 21-26	• Gospel Reading, Reflection, and Prayer Discussion of Module 5 – System Modeling EA: Laboratory Activity 6: Submission of the discussion of the Modeling used in their study	MS Teams PowerPoint Schoolbook	3
	Week 11 Oct 28 -Nov 2	***Independent Learning Week***		
	Week 12 Nov 4-9	• Gospel Reading • Continuation of Module 5 - System Modeling EA: Lab Activity 7: • Submission of the discussion of the Algorithms used in the Group Project and of the discussion of Operational Framework and Use Cases, Flowchart, State Diagram for the Group Project.	MS Teams PowerPoint Schoolbook	3
	Week 13 Nov 11-16	• Gospel Reading, Reflection, and Prayer • Project Development Time	MS Teams PowerPoint Schoolbook	3
Module 6 Module 7 TLO 22 TLO 23 TLO 24 TLO 25 TLO 26 TLO 27 TLO 28 TLO 29 CLO 6 CLO 7 CLO 8	Week 14 Nov18-23	• Gospel Reading, Reflection, and Prayer • Online Project Consultation • EA: Initial Submission of Project (Software and Documentation)	Schoolbook	3
	Week 15 Nov 25-30	• Gospel Reading, Reflection, and Prayer • Project Development Time	PowerPoint Schoolbook	3
	Week 16 Dec 2-7	• Gospel Reading, Reflection, and Prayer • EA: Final Submission of Project (Software and Documentation)	MS Teams	3

	Week 17 Dec 9-14	• Gospel Reading, Reflection, and Prayer • Summative Assessment 2: Project Defense / Presentation	PowerPoint Schoolbook	3
TOTAL				24

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	40%	Enabling Assessments	40%
Summative Assessment	60%	Summative Assessment	60%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies

a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.
d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback

a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:

i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.

ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.

iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES:

On-Site References

Provider	Reference Material
AEA Library QA 76.758 .L465 2016	Leach, RJ. 2016. Introduction to Software Engineering. 2nd Edition. (Boca Raton): CRC Press, Taylor & Francis Group
AEA Library QA 76.758 .P926 2015	Pressman, RS. 2015 . Software engineering: a practitioner's approach. (Singapore): McGraw-Hill Book Co.

Online References

Call Number/ E-provider	Reference Material
Web Resource	Alliance Software. 2020. An Introduction to Software Development Methodologies. [accessed 2024 July 30] https://www.alliancesoftware.com.au/introduction-software-development-methodologies/
Web Resource	Muhammad Raza. 2020. The Software Development Life Cycle: An Introduction. [accessed 2024 July 30] https://www.bmc.com/blogs/sdlc-software-development-lifecycle/
Web Resource	Carriou Erwan. 2019. Designing System. [accessed 2024 July 30] https://medium.com/swlh/designingsystemsce28ee31c0cd

Web Resource	Study.com 2021. Video Team: Requirements Modelling in Software Engineering. [accessed 2024 July 30] https://study.com/academy/lesson/requirements-modeling-in-software-engineering-classesfunctionsbehaviors.html
Web Resource	Guru99. 2020. What is System Testing? Types and Definition. [accessed 2024 July 30] https://www.guru99.com/system-testing.html

Prepared: **Information Technology Department**
1st Semester 2024-2025

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